

stress, fear, mental effort, and physical exertion.³² Making use of such features in cognitive systems involves:

- developing a physicon that maps sensor outputs to ontologically grounded feature values;
- developing a dedicated recognition module that perceives physiological inputs and interprets them according to the physiocon;
- developing a dedicated interpretation module that contextually interprets physiological features in terms of people's physical, emotional, and cognitive states: for example, in a given context, increased heart rate might be explained by stress, physical exertion, or exposure to heat; and
- developing reasoning functions that guide the agent in responding to different human states depending on a large number of features of the context including the respective roles that the human and agent are playing, the type of application, the application domain, and so on.

Returning to the overall topic of this section, just as LEIAs need a lexicon to map between language and ontologically grounded meanings, they need analogous knowledge bases to map between other channels of input and ontologically grounded meanings. It is these meanings that agents use as input for reasoning about action.

3.5 Episodic Memory

Episodic memory includes stored information about instances (exemplars) of ontological concepts as well as meaning representations that the agent generates during its operation. Episodic knowledge structures include time-stamps, provenance, and other relevant metadata.

The agent's episodic memory is divided into spaces, which is a common practice in memory management. Memory spaces allow the agent to rapidly access sets of known instances that share a common category or purpose. For example, one part of an agent's episodic memory contains a fact repository of the agent's beliefs about entities it knows—for example, the capital of Belgium and the hair color of its human collaborator Ben. Another part of the agent's episodic memory contains information about its past successes and failures at carrying out a particular kind of plan (recall that plans are instances of ontological scripts). Memories of plans allow the agent to, in certain cases, bypass detailed decision-making about action

and, instead, carry out a reflexive action. Specifically, when the agent needs to decide which plan to use to achieve a goal, it can search its episodic memory for past cases when particular plans successfully achieved the goal. It can then compare its current situation model to the situation models associated with the successful plans and select the best match. The agent can then instantiate another copy of that plan in anticipation that it will work as well as the last time. This kind of operation is an example of case-based reasoning.

The content of the agent's episodic memory is made available to all reasoning heuristics throughout the system, allowing any algorithm to inspect what the agent knows, what it has recently encountered, what it is currently thinking about, and what is on its agenda. As with ontological knowledge, episodic memory is indexed in a variety of ways—by relevant domain space, but also by type, timestamp, and more. Operations devoted to consolidation and other updates of the episodic memory, as well as the way the agents model forgetting, are outside the scope of this book.